Chrystal Chern

Dynamics Lab / STAIRLab / Berkeley Discovery
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Dynamics researcher specializing in IMU-based motion sensing, system identification, and digital twins of physical systems, with experience spanning sensor instrumentation, standards-based experimental testing, and signal-level characterization through production-quality Python analysis tools.

Education

2019–2024 **PhD | Structural Engineering, Mechanics, and Materials**, *University of California*, Berkeley, GPA: 3.89/4.00.
Thesis: *Digital Twin Framework for Vibration-Based Structural Health Monitoring*
Advisor: Khalid M. Mosalam
Major Focus: Structural mechanics and dynamics, linear and nonlinear structural analysis, finite element methods, performance-based engineering and design.
Minor Focus: Artificial intelligence, machine learning, control systems, and scientific computing.

2019 **MS | Structural Engineering, Mechanics, and Materials**, *University of California*, Berkeley.
Thesis: *Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets*

2016 **BS | Mechanical Engineering**, *Massachusetts Institute of Technology*, Cambridge.
Mechanical systems, thermodynamics, and design.

Research Experience

University of California, Berkeley

2024–Present **Postdoctoral Researcher**, *Dynamics Lab & STAIRLab*, Berkeley, CA.
Dynamics Lab:
- Lead instrumentation, signal processing, and inverse dynamic analysis of coupled human-object biomechanical systems using multi-node IMU sensor networks with strap-down integration and magnetic field mapping; led experiment design, sensor placement, and data collection.
- Developed network analysis methodology and Python analysis code to identify the causal body-segment motions driving induced object motion (*Nonlinear Dynamics*, submitted 2026).

STAIRLab (STructural Artificial Intelligence Research Lab):
- Build finite element models to generate analytical benchmarks, validating system-identification methods across experimental, field, and FEA-simulated structural response data.
- Evaluate system identification prediction residuals and modal clustering for anomaly detection and early-warning structural health monitoring under environmental and seismic loads.
- Supervise graduate and undergraduate researchers in data-driven system dynamics.

2019–2024 **Graduate Student Researcher**, *STAIRLab & Pacific Earthquake Engineering Research Center (PEER)*, Berkeley, CA.
Bridge Rapid Assessment Center for Extreme Events (BRACE2):
- Developed California's production web platform for automated bridge health monitoring (Caltrans).
- Performed system identification and finite element modeling of bridges and buildings.

Structural ImageNet & Feature Engineering for Structural Health Monitoring:
- Developed machine learning anomaly detection for bridge damage diagnosis within a human-machine collaboration framework (*Journal of Bridge Engineering*, 2024).
- Developed deep learning image classification models for structural health monitoring.
- Built feature engineering pipelines for damage classification.

Massachusetts Institute of Technology

2015–2016 **Undergraduate Researcher**, *MIT Digital Structures Lab*, Cambridge, MA.
Experimental characterization of fine-resolution anisotropic material properties of 3D printed materials via ASTM tensile testing.

Professional Engineering Experience

2016–2018 **Staff I – Building Technology**, *Simpson Gumpertz & Heger Inc.*, Washington, DC.

- Performed parametric energy, daylighting, and thermal finite element analysis (2D/3D) of curtain wall, cladding, and custom skylight assemblies.
- Conducted infrared thermography surveys and analysis for non-destructive diagnostics.
- Performed on-site water penetration and air infiltration testing of installed cladding assemblies per ASTM, AAMA, and ASHRAE standards, including instrumented pressure-chamber test setup, acceptance-criteria evaluation, and leak-path diagnosis.
- Analyzed wind loads on components and cladding per structural code.

2014 **Engineering Intern**, *Dassault Systemes SOLIDWORKS Corp.*, Woodland Hills, CA.

- Investigated user-reported software issues across Computer Aided Design (CAD) and Finite Element Analysis (FEA) products for the SOLIDWORKS 2015 Beta program.
- Contributed to the redesign of the engineering technical support knowledge base.

Technical & Computer Skills

Programming (production)	Python (package/module development), Bash, Tcl, HTML/CSS, JavaScript
Additional Languages & Tools	MATLAB, R, C, Rust
Modeling & Analysis	Vibrations and structural dynamics, differential equation modeling, linear algebra, numerical methods, finite element analysis of material and structural behavior (mechanical and thermal), machine learning anomaly detection
Test & Measurement	Data acquisition (DAQ), sensor instrumentation, signal processing (FFT, Kalman filtering, strap-down integration, system identification), magnetic field mapping, ASTM standards-based testing (tensile, water penetration, air infiltration), infrared thermography, field acceptance testing
Languages	English (Native), Mandarin Chinese, Spanish

Fellowships & Awards

2024–Present **UC Berkeley Future of Higher Education Postdoctoral Fellowship** ($140k)

2020–2024 **NSF Graduate Research Fellowship (GRFP)** ($160k)

2015 **MIT Priscilla King Gray Public Service Center Expedition Grant** ($10k)

2012 **Intel Scholarship** ($4k)

Publications

Journal Articles

In Press **Chrystal Chern** and Oliver M. O’Reilly. “On Complex Modal Inverse Analysis: The Cautionary Tale of the Single Surviving Mode.” *ASME Journal of Applied Mechanics*. Accepted July 2026.

Submitted **Chrystal Chern**, Theresa E. Honein, and Oliver M. O'Reilly. "The Symphony of Gyrations Producing Steady Motions of a Hula Hoop: Insights from a Network Analysis." *Nonlinear Dynamics*. Submitted May 2026.

In Prep. **Chrystal Chern** and Khalid Mosalam. "Modal Clustering for Digital Twinning of Civil Infrastructure." *Journal of Engineering Mechanics*.

In Prep. Naiqi Guo, **Chrystal Chern**, and Khalid Mosalam. "Detecting Inelasticity in Seismic Data with Time Response System Identification." *Mechanical Systems and Signal Processing*.

2024 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *Journal of Bridge Engineering*, 29(7), 04024041.

Conference Proceedings

2025 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "Structural Response Prediction from Learned System Realization Matrices." *11th International Conference on Experimental Vibration Analysis of Civil Engineering Structures (EVACES)*.

2024 **Chrystal Chern** and Khalid Mosalam. "Cross-Sectional Study of Physics-Informed Bridge Health Identification." *International Association of Bridge Earthquake Engineering (IABEE) Fourth International Bridge Seismic Workshop (4IBSW)*.

2020 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *SMIP20 Seminar on Utilization of Strong-Motion Data Proceedings*, pp. 100-127.

Software

2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. `mdof: 0.0.16-alpha`, open-source package for system identification. Zenodo. `https://doi.org/10.5281/zenodo.11660201`

Technical Reports & Dissertations

2024 **Chrystal Chern**. "Digital Twin Framework for Vibration-Based Structural Health Monitoring." *PhD Dissertation*, UC Berkeley.

2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "BRACE2: Bridge rapid assessment center for extreme events, Phase I Final Report." *Caltrans Technical Report No. CA24-3703*.

2019 **Chrystal Chern**. "Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets." *MS Research Report*.

Teaching & Mentorship

Spring 2026 **Lecturer (Instructor of Record)**, *L&S 110 – Brilliance of Berkeley (UC Berkeley)*.
Advise students, prepare course materials, grade, facilitate discussions, and coordinate guest speakers; supervise student readers.

2024–2025 **Teaching Assistant**, *MAS-E ENGIN 235-A – Python for Engineers (UC Berkeley)*.
Authored programming laboratory assignments, developed autograder scripts, and reviewed lectures and quizzes.

Dec 2024 **Guest Lecturer**, *CE 249 – Experimental Methods in Structural Engineering (UC Berkeley)*.
Delivered a lecture on system identification of vibration data using computer vision.

2019, 2023–2024 **Graduate Student Instructor**, *E7 – Computer Programming for Scientists and Engineers (UC Berkeley)*.
Delivered discussion lectures, worksheets, and laboratory assignments; led exam review sessions and graded exams.

2025 **Lead Organizer & Founder**, *SEMM Graduate Program Primer Boot Camp*, UC Berkeley.
Founded and taught a primer on core theory and field-specific conventions for incoming graduate students; organized 7 doctoral co-instructors.

Service & Leadership

2024–Present **Program Strategist**, Berkeley Discovery Initiative, UC Berkeley – Co-developed an interdisciplinary research mentorship program (8 research hubs, 30+ graduate fellows, 170 undergraduates); led survey and focus-group assessment and co-secured $170k in continuing funding.

2022–Present **Technical Reviewer** for *Resilient Cities and Structures*, *Computers and Structures*, and *Computer-Aided Civil and Infrastructure Engineering*.

2024–Present **Research Mentor**, UC Berkeley SEMM & CEE Scholars Programs – Designed and supervised structural health monitoring and sensor projects for 5 undergraduate researchers.

2022–2024 **Founder & President**, UC Berkeley SEMM-DSR (Doctoral Students and Researchers Organization).

2014–2015 **Malawi Project Manager & Fundraising Chair**, MIT Engineers Without Borders.

References

Oliver M. O'Reilly
Postdoctoral Research Supervisor
Distinguished Professor of Mechanical Engineering, UC Berkeley
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Khalid M. Mosalam
PhD Thesis Advisor & Postdoctoral Supervisor
Taisei Professor of Civil Engineering & Director, PEER Center, UC Berkeley
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